# Job details

$\sin^2(\theta_{12}) = 0.304$

$\sin^2(\theta_{13}) = 0.0217$

$\sin^2(\theta_{23}) = 0.528$

$\Delta m^2_{21} = 7.53\text{e-}05$

$|\Delta m^2_{32}| = 0.002509$

$\delta_{CP} = 3.141592654$

BANFF tunings: Enabled
Hierarchy: Normal

# 1Rmu

| | $\nu_\mu$ | $\nu_e$ | $\bar{\nu}_\mu$ | $\bar{\nu}_e$ | Osc. $\nu_e$ | Osc. $\bar{\nu}_e$ | Total |
|---|---|---|---|---|---|---|---|
| CCQE | 171.79503 | 0.00398 | 10.84413 | 0.00019 | 0.01680 | 0.00019 | 182.66032 |
| CC1pi | 27.08913 | 0.00195 | 2.60005 | 0.00009 | 0.01902 | 0.00009 | 29.71032 |
| CCcoherent | 0.30254 | 0.00000 | 0.09754 | 0.00000 | 0.00000 | 0.00000 | 0.40009 |
| 2p2h | 33.44538 | 0.00144 | 0.60206 | 0.00003 | 0.01842 | 0.00002 | 34.06734 |
| CCother | 5.70373 | 0.00093 | 0.42707 | 0.00008 | 0.00009 | 0.00001 | 6.13191 |
| NC1pi0 | 0.59683 | 0.01773 | 0.02060 | 0.00137 | N/A | N/A | 0.63654 |
| NC1piPM | 4.74956 | 0.09356 | 0.16977 | 0.00935 | N/A | N/A | 5.02223 |
| NCcoherent | 0.00000 | 0.00000 | 0.00044 | 0.00003 | N/A | N/A | 0.00047 |
| NCother | 2.05462 | 0.07356 | 0.13136 | 0.00914 | N/A | N/A | 2.26867 |
| Total | 245.73682 | 0.19315 | 14.89302 | 0.02028 | 0.05432 | 0.00031 | 260.89790 |

Total: 260.89790 (260.89790)
Total signal: 245.73682
Total background: 15.16108
POT: 14.73410E20

# 1Re

| | $\nu_\mu$ | $\nu_e$ | $\bar{\nu}_\mu$ | $\bar{\nu}_e$ | Osc. $\nu_e$ | Osc. $\bar{\nu}_e$ | Total |
|---|---|---|---|---|---|---|---|
| CCQE | 0.15697 | 6.10636 | 0.00629 | 0.24827 | 35.96660 | 0.35330 | 42.83779 |
| CC1pi | 0.04638 | 0.90811 | 0.00278 | 0.06685 | 3.45502 | 0.05309 | 4.53224 |
| CCcoherent | 0.00012 | 0.00867 | 0.00019 | 0.00414 | 0.02828 | 0.00420 | 0.04561 |
| 2p2h | 0.02876 | 1.48627 | 0.00050 | 0.01793 | 6.41380 | 0.02026 | 7.96752 |
| CCother | 0.01405 | 0.13612 | 0.00036 | 0.01001 | 0.07370 | 0.00348 | 0.23772 |
| NC1pi0 | 1.65527 | 0.03550 | 0.05865 | 0.00344 | N/A | N/A | 1.75285 |
| NC1piPM | 0.16122 | 0.00444 | 0.00975 | 0.00054 | N/A | N/A | 0.17595 |
| NCcoherent | 0.52967 | 0.00714 | 0.04844 | 0.00294 | N/A | N/A | 0.58819 |
| NCother | 0.31724 | 0.01691 | 0.02133 | 0.00128 | N/A | N/A | 0.35676 |
| NC1gamma | 1.88841 | 0.03365 | 0.10053 | 0.00467 | N/A | N/A | 2.02725 |
| Total | 4.79809 | 8.74317 | 0.24883 | 0.36006 | 45.93740 | 0.43433 | 60.52188 |

Total: 60.52188 (60.52188)
Total signal: 45.93740
Total background: 14.58448
POT: 14.73410E20

## RHC1Rmu

| | $\nu_\mu$ | $\nu_e$ | $\bar{\nu}_\mu$ | $\bar{\nu}_e$ | Osc. $\nu_e$ | Osc. $\bar{\nu}_e$ | Total |
|---|---|---|---|---|---|---|---|
| CCQE | 15.20209 | 0.00083 | 28.97808 | 0.00090 | 0.00059 | 0.00177 | 44.18426 |
| CC1pi | 3.88925 | 0.00032 | 4.46367 | 0.00025 | 0.00048 | 0.00079 | 8.35475 |
| CCcoherent | 0.04974 | 0.00000 | 0.18383 | 0.00005 | 0.00000 | 0.00009 | 0.23371 |
| 2p2h | 4.22962 | 0.00024 | 1.20133 | 0.00011 | 0.00033 | 0.00012 | 5.43176 |
| CCother | 0.99428 | 0.00012 | 0.62888 | 0.00000 | 0.00000 | 0.00000 | 1.62328 |
| NC1pi0 | 0.05808 | 0.00213 | 0.05953 | 0.00163 | N/A | N/A | 0.12137 |
| NC1piPM | 0.34687 | 0.01454 | 0.44406 | 0.01183 | N/A | N/A | 0.81730 |
| NCcoherent | 0.00000 | 0.00000 | 0.00202 | 0.00000 | N/A | N/A | 0.00202 |
| NCother | 0.31819 | 0.01642 | 0.19923 | 0.00786 | N/A | N/A | 0.54170 |
| Total | 25.08812 | 0.03460 | 36.16062 | 0.02264 | 0.00140 | 0.00276 | 61.31014 |

Total: 61.31014 (61.31014)
Total signal: 36.16062
Total background: 25.14952
POT: 7.55730E20

## RHC1Re

| | $\nu_\mu$ | $\nu_e$ | $\bar{\nu}_\mu$ | $\bar{\nu}_e$ | Osc. $\nu_e$ | Osc. $\bar{\nu}_e$ | Total |
|---|---|---|---|---|---|---|---|
| CCQE | 0.01262 | 0.49310 | 0.02343 | 0.82190 | 0.79433 | 3.78097 | 5.92635 |
| CC1pi | 0.00755 | 0.09834 | 0.00724 | 0.14292 | 0.10922 | 0.41234 | 0.77761 |
| CCcoherent | 0.00003 | 0.00104 | 0.00044 | 0.01114 | 0.00094 | 0.04331 | 0.05689 |
| 2p2h | 0.00281 | 0.13374 | 0.00093 | 0.04739 | 0.17164 | 0.18567 | 0.54218 |
| CCother | 0.00416 | 0.02249 | 0.00126 | 0.01227 | 0.00868 | 0.00748 | 0.05633 |
| NC1pi0 | 0.13495 | 0.00520 | 0.17142 | 0.00451 | N/A | N/A | 0.31607 |
| NC1piPM | 0.02361 | 0.00121 | 0.01922 | 0.00077 | N/A | N/A | 0.04480 |
| NCcoherent | 0.04822 | 0.00221 | 0.20250 | 0.00317 | N/A | N/A | 0.25610 |
| NCother | 0.05700 | 0.00221 | 0.02653 | 0.00097 | N/A | N/A | 0.08672 |
| NC1gamma | 0.16043 | 0.00878 | 0.34899 | 0.00763 | N/A | N/A | 0.52582 |
| Total | 0.45138 | 0.76831 | 0.80194 | 1.05266 | 1.08481 | 4.42977 | 8.58887 |

Total: 8.58887 (8.58887)
Total signal: 4.42977
Total background: 4.15911
POT: 7.55730E20

## MultiRe

| | $\nu_\mu$ | $\nu_e$ | $\bar{\nu}_\mu$ | $\bar{\nu}_e$ | Osc. $\nu_e$ | Osc. $\bar{\nu}_e$ | Total |
|---|---|---|---|---|---|---|---|
| CCQE | 0.02764 | 0.02365 | 0.00114 | 0.00168 | 0.16283 | 0.00363 | 0.22058 |
| CC1pi | 0.07500 | 0.74437 | 0.00259 | 0.00379 | 3.74861 | 0.00253 | 4.57688 |
| CCcoherent | 0.00029 | 0.01846 | 0.00008 | 0.00007 | 0.09323 | 0.00012 | 0.11227 |
| 2p2h | 0.00963 | 0.02048 | 0.00021 | 0.00040 | 0.11602 | 0.00052 | 0.14725 |
| CCother | 0.04103 | 0.12014 | 0.00126 | 0.00298 | 0.10172 | 0.00167 | 0.26880 |
| NC1pi0 | 0.01970 | 0.00072 | 0.00081 | 0.00008 | N/A | N/A | 0.02130 |
| NC1piPM | 0.06066 | 0.00157 | 0.00367 | 0.00031 | N/A | N/A | 0.06621 |
| NCother | 0.20532 | 0.00828 | 0.01312 | 0.00089 | N/A | N/A | 0.22761 |
| NC1gamma | 0.05689 | 0.00000 | 0.00135 | 0.00014 | N/A | N/A | 0.05838 |
| Total | 0.49616 | 0.93766 | 0.02422 | 0.01034 | 4.22242 | 0.00848 | 5.69928 |

Total: 5.69928 (5.69928)
Total signal: 4.22242
Total background: 1.47686
POT: 14.73410E20